Renuka Jagadale

Data Scientist | Machine Learning Engineer | AI Engineer | Software Engineer

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SUMMARY

Available starting May 2025. Data Science and Machine Learning professional with 7 years of experience in the software development, specializing in data analysis, predictive modeling, and algorithm development.

EDUCATION

Pursuing Master of Science (Computer Science) 08/2024 – present | Southfield, USA
Lawrence Technological University
Coursework: Machine Learning, Deep Learning, GPA 3.9.

Bachelor of Engineering, Pune University 08/2012 – 06/2016 | Pune, India
Graduated with 3.52 GPA on scale of 4.

TECHNICAL SKILLS :

- **Machine Learning**: Linear Regression, Logistic Regression, KNN, Naive Bayes, SVM, Decision tree, Random Forest.
- **Deep Learning**: Artificial Neural Networks, CNN, LSTM, RNN, Seq2seq models, LLMs.
- **Libraries and Modules**: Pandas, Numpy, Seaborn, NLTK, Spacy, Matplotlib, Scikit-learn, Keras, Tensorflow.
- **Language**: Python, SQL
- **Deploymen**t: Flask, Heroku.
- **Visualization:** Power BI

WORK EXPERIENCE

Data Scientist 03/2023 – 08/2024 | Pune, India
Flex (Flextronics Technologies India Private Limited)

Project: Forecasting Production Time:
- Developed a machine learning model using H2O AutoML to forecast production time for various parts on the manufacturing line. It helped to optimize production scheduling by accurately predicting the time required to manufacture parts.
- Collected and preprocessed historical production data, incorporating features such as part specifications, machine type, and operator shifts.
- Evaluated multiple regression models and selected the best-performing one based on RMSE and cross-validation metrics.
- Integrated the forecasting model into the production planning system to enable real-time task scheduling.
- It improved scheduling accuracy and resource allocation, resulting in a 20% reduction in production delays and increased operational efficiency.

Project: Automatic Clause Extraction on Legal Contracts Using GenAI:
- Built a tool that reads and processes legal contracts in PDF format using specialized document-reading tools in Python.
- Used Generative AI to automatically extract important legal sections such as payment terms, warranty conditions, and currency exchange details from agreements between suppliers and customers.
- Designed prompts and logic to make sure the AI captures clauses accurately and in the right context.
- Structured the extracted information so it could be easily reviewed or used by legal and business teams.
- It reduced the time and manual effort required to review legal contracts by 70%, enabling faster decision-making and minimizing the risk of missed or incorrect terms.

Project: Document Search Engine:
- To simplify access to key financial insights by enabling natural language search across complex documents, developed a smart document search tool that allows users to ask questions and retrieve answers directly from large financial reports.
- Used Large Language Models (LLMs) (similar to ChatGPT) to understand user queries and extract relevant information from various finance-related documents.
- Designed the system to be reusable and scalable, allowing it to handle different types of documents across departments.
- Improved document accessibility for non-technical users by allowing them to get answers in plain language without manually reading lengthy files.
- It significantly reduced time spent on reviewing reports, improved team productivity, and enabled faster data-driven decisions.

Data Miner (Data scientist) 07/2021 – 03/2023 | Bangalore, India
Concentrix Daksh Services Private Limited

Project: Hierarchical Topic Classification on Survey Data:

- To uncover key themes and sentiments from customer feedback to improve service quality, analyzed survey data using KPIs like NPS and CSAT to generate actionable insights.
- Applied NLP techniques (tokenization, tagging, sentiment analysis) to understand customer opinions.
- Built a dictionary-based model to classify topics into categories like product, service, pricing, and organization.
- Created an interactive Power BI dashboard to visualize insights.
- It helped boost CSAT score from 6.5 to 9 by identifying and addressing key customer concerns.

Project: Social Media Analysis on Twitter Data:

- To compare brand perception and customer satisfaction using Twitter data, collected and analyzed tweets for two competing brands using web scraping.
- Performed sentiment analysis to evaluate customer opinions and satisfaction levels.
- Built a classification model to group tweets by topics such as product, service, or support.
- Visualized key insights using Power BI dashboards.
- It improved understanding of brand perception and uncovered opportunities to enhance customer service.

Data Analyst 12/2018 – 04/2021 | Pune, India
Explicate Technologies Private Limited

Project: Customer Insights & Sales Optimization in Hospitality & Retail

- Analyzed customer behavior and booking data to identify trends and optimize marketing and pricing strategies to boost sales and enhance customer engagement through data-driven strategies.
- Built pivot tables and visual dashboards to support strategic decision-making.
- Applied clustering techniques for personalized marketing and loyalty program design.
- Conducted sentiment analysis and data cleaning to improve customer experience.
- This enabled dynamic pricing and improved targeting, leading to a 15% increase in sales.

Android UI Developer 12/2017 – 12/2018 | Pune, India
Explicate Technologies Private Limited

- Designed user interfaces like appropriate icons and display screens for Android applications using XML.
- Integrated Google analytics in Android application to collect user activity.
- Created updated themes, animations, and layouts.
- Wrote technical specifications and maintained all the reports of the application development process.
- Tested the application on different Android targets and debugged the issues related to the user interface.

PROJECTS

Interpretable Chatbot using Inductive Logic Programming and Machine Learning 04/2025 – Present

Objective: To develop a transparent and explainable chatbot for domain-specific applications by combining symbolic reasoning with machine learning.

Description:

- Built a hybrid chatbot system that integrates Inductive Logic Programming (ILP) with ML models (Decision Trees, Random Forests) for interpretable intent detection.
- Transformed dialogue data into structured logic rules and predicate forms (e.g., intent, slot-value pairs) using ILP.
- Engineered additional features like question-type detection, bigrams, and turn position to enhance classification accuracy.
- Achieved up to 90% accuracy while maintaining full transparency and debuggability, addressing the limitations of black-box NLP models.
- Delivered an explainable and maintainable alternative to traditional chatbot models, ideal for business-critical and regulated domains.

COURSES AND CERTIFICATIONS

Applied Machine Learning Course
Applied AI

Completed a Machine Learning course focusing on data analysis using Python, covering various algorithms including supervised and unsupervised learning, and deep learning techniques such as Neural Networks, CNN, LSTM, and Transformers. Gained hands-on experience in model training, evaluation, and optimisation using tools like Python, TensorFlow, and Scikit-learn.